AAAI 2026 FALL SYMPOSIUM

Some Lessons from *Smart Transportation* and Beyond to Help *Plan for a Better Planet*



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University of South Carolina

Keywords: *Smart Transportation, Sustainability, AI, Planning, Lessons, Climate Change*

AI INSTITUTE #AIISC UNIVERSITY OF SOUTH CAROLINA

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AAAI 2026 Fall Symposium on **Planning for a Better Planet**

ROADMAP

Outline

01	Trends: Technology for a Better Planet	• System of system inter-dependency • Socio-economic-technical problems
02	Context: Initiatives in Smart Transportation and Traffic	
03	Lessons	
04	Discussion	• Beyond transportation • Planning research

01

Trends: Technology for a Better Planet

Major Community Challenges

CHALLENGES

- **Growing population**
  - Access to safe water
  - Need jobs
- **Reducing resources**
  - Land
  - Potable water
- **Environmental degradation**
  - Air pollution
  - Animal management



South Carolina's traffic deaths rose by 26 percent from 1994 to 2020, **marking the biggest increase by any state over those 26 years**. **In 2022, 1,035 people died on SC roads**, per preliminary data reported by the SC Department of Public Safety

Credit: https://www.greenvilleonline.com/story/news/2022/12/29/fatalities-on-sc-roads-decline-in-2022-greenville-roads-dangerous-spartanburg-anderson-traffic/69763182007/

Attempts at Solving

Computational Sustainability

Research community led

Smart City

IT industry led
(IBM, Cisco, …)

UN’s Millenium Development Goals

Governance led

01 • TRENDS

System of System Problems

A system (dimension) of the problem is connected with another

For solving traffic (transportation),

- **Needed:** energy
- **Impacts:** air (pollution), health

For solving energy,

- **Needed:** water, transportation
- **Impacts:** air (pollution), health

For solving water,

- **Needed:** energy
- **Impacts:** health, land

...

No system's problem can be solved fully in isolation; ***watch-out for long-term effects***

Goals: Smart City? Millennium Development Goals?

Smart city can mean one or more of the following:

As a **resource optimization objective**, it is to know and manage a city's resources using data.

As a **caring objective**, it is about improving standard of life of citizens with health, safety, etc indices and programs.

As a **vitality objective**, it is about generating employment and doing sustainable growth.

A city leadership can choose among these or define their own objective(s) and manage with measurements to pro-actively achieve it



See other FAQs at: https://sites.google.com/site/biplavsrivastava/research-1/intelligent-systems/scfaqs

Credits: See: https://www.un.org/millenniumgoals/bkgd.shtml

01 • TRENDS

Major Community Challenges and Opportunities Today

CHALLENGES

- **Growing population**
  - Access to safe water
  - Need jobs
- **Reducing resources**
  - Land
  - Potable water
- **Environmental degradation**
  - Air pollution
  - Animal management

AI OPPORTUNITIES

- **Trends**
  - Lots of data
  - Cheap compute
  - Access to online information
- **Many avenues for impact**
  - Water management – e.g., clean plastics [1]
  - Disease – detecting from photos [2]
  - Video-based animal management
  - Intelligent traffic control
  - Promoting sustainable tourism
  - Journalism - Media coverage of homicides and role of race [2]

Vision: create participative and trustable technology solving real-world problems

2015 IJCAI Tutorial - **AI for Smart City Innovations with Open Data**
https://www.linkedin.com/pulse/tutorial-ai-smart-city-innovations-open-data-biplav-srivastava/

References:
1. https://theconversation.com/new-pfas-guidelines-a-water-quality-scientist-explains-technology-and-investment-needed-to-get-forever-chemicals-out-of-us-drinking-water-201855
2. https://www.prnewswire.com/news-releases/students-win-more-than-1-8-million-at-2023-regeneron-science-talent-search-for-remarkable-scientific-research-on-rna-molecule-structure-media-bias-and-diagnostics-for-pediatric-heart-disease-301772440.html

01 • TRENDS

Real-World Applications of Technology Follow a Pattern

Value (from Action, Decisions)

- Providing benefits that matter to people most in need of, and in a timely and cost-efficient manner.
- Going beyond technology to process and people aspects.

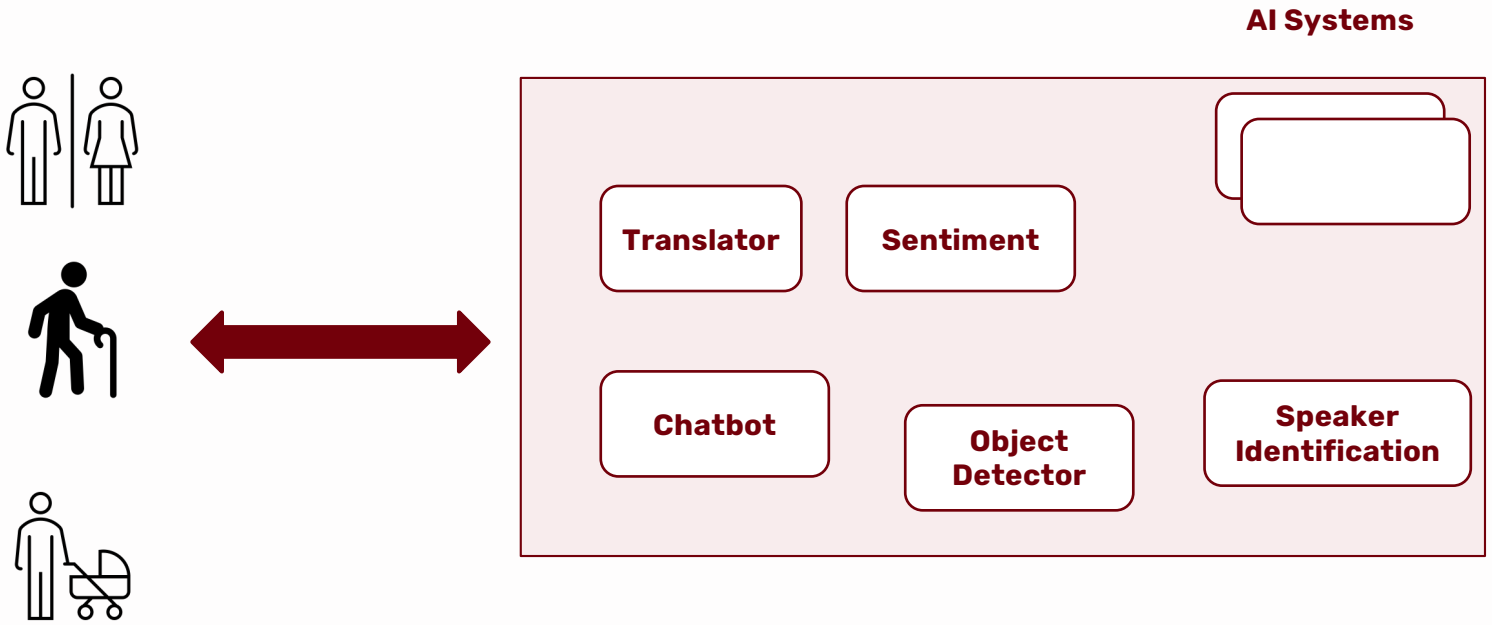
Data + Insights

- Available,
- Consumable with Semantics,
- Analysis, often including Visualization too

Access

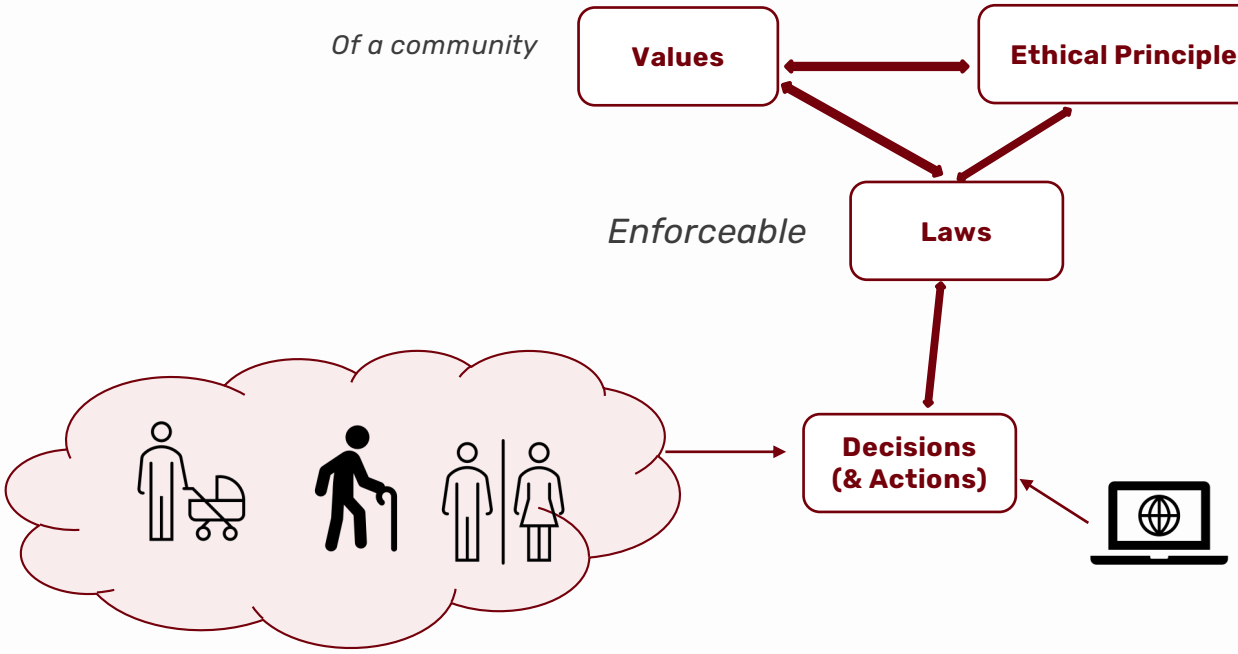
- Apps (Applications),
- Usability - Human Computer Interface,
- Application Programming Interfaces (APIs)

Technology and People



Trust: *Can people trust AI systems to perform capably, consistently, and with human values?*

Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can, humans or machines (AI) or in collaboration together, take decisions and actions following the same **values**, **ethical principles** and **laws** ?*

Assumption: *they should follow same values, principles and laws.*

02

Context: Initiatives in Smart Transportation and Traffic

02 • CONTEXT

Many Dimensions

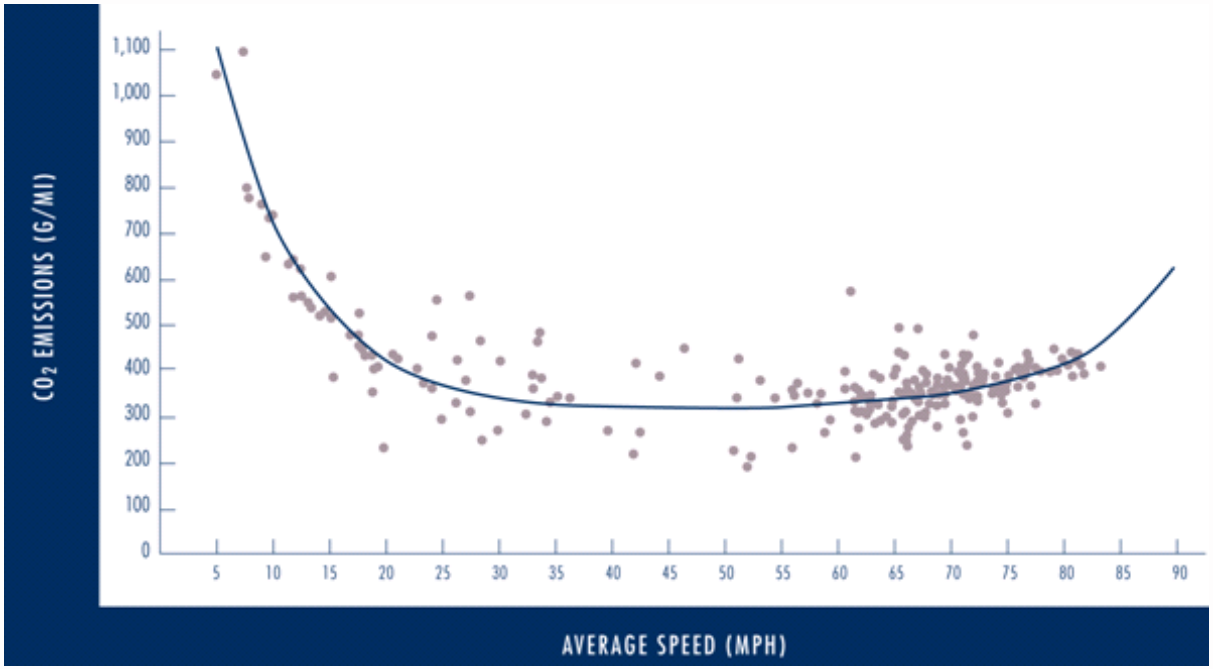
Understanding the problem

Data collection to sense traffic

Data analysis for specific **technical** goal (e.g., traffic intervention)

Using multiple analyses together for **business** goal (improving traffic safety)

Relationship Between Congestion and Greenhouse Gases



- Very low and very high traffic speeds lead to higher emissions
- Moderate speed has low emissions

Source: Traffic Congestion and Greenhouse Gases, by Matthew Barth and Kanok Boriboonsomsin
From: http://www.uctc.net/access/35/access35_Traffic_Congestion_and_Grenhouse_Gases.shtml

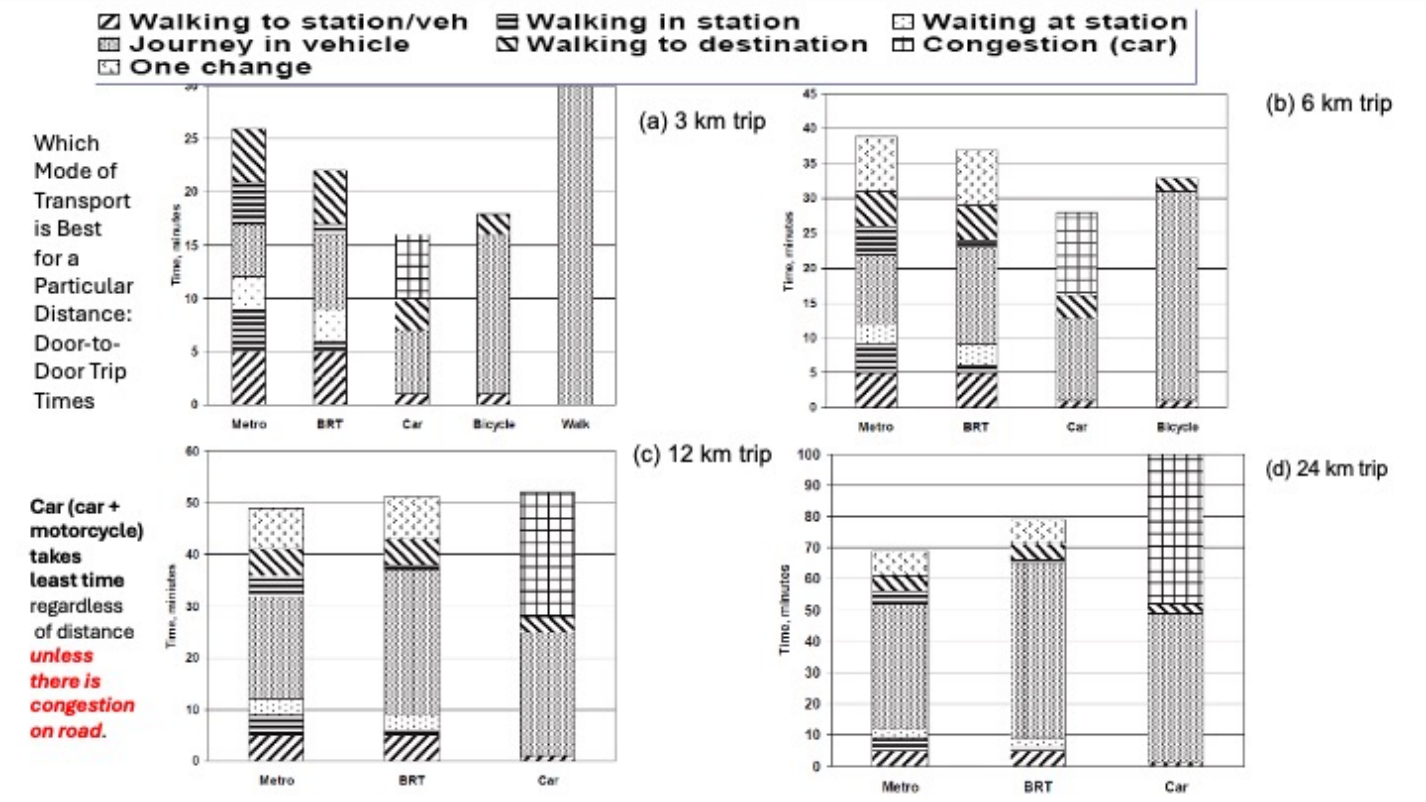
Understanding Who Wants to Move and When

Mode	Non-Drivers	Poor	Handi-capped	Limitations	Most Appropriate Uses
Walking	Yes	Yes	Varies	Requires physical ability. Limited distance and carrying capacity. Difficult or unsafe in some areas.	Short trips by physically able people.
Wheelchair	Yes	Yes	Yes	Requires sidewalk or path. Limited distance and carrying capacity.	Short urban trips by people with physical disabilities.
Bicycle	Yes	Yes	Varies	Requires bicycle and physical ability. Limited distance and carrying capacity.	Short to medium length trips by physically able people on suitable routes.
Taxi	Yes	Limited	Yes	Relatively high cost per mile.	Infrequent trips, short and medium distance trips.
Fixed Route Transit	Yes	Yes	Yes	Destinations and times limited.	Short to medium distance trips along busy corridors.
Paratransit	Yes	Yes	Yes	High cost and limited service.	Travel for disabled people.
Auto driver	No	Limited	Varies	Requires driving ability and automobile. High fixed costs.	Travel by people who can drive and afford an automobile.
Ridesharing (auto passenger)	Yes	Yes	Yes	Requires cooperative automobile driver. Consumes driver's time if a special trip (chauffeuring).	Trips that the driver would take anyway (ridesharing). Occasional special trips (chauffeuring).
Carsharing (Vehicle Rentals)	No	Limited	Varies	Requires convenient and affordable vehicle rentals services.	Occasional use by drivers who don't own an automobile.
Motorcycle	No	Limited	No	Requires riding ability and motorcycle. High fixed costs.	Travel by people who can ride and afford a motorcycle.
Telecommute	Yes	Varies	Varies	Requires equipment and skill.	Alternative to some types of trips.

Each mode is suitable for certain types of travel. None is a perfect substitute for driving.

Source: http://www.vtpi.org/tdm/tdm103.htm

Understanding the Traffic Problem



02.1

Data Analysis Case Study – SCC Boston (2012)

SCC Boston Team and Press Coverage



SCC Boston team with Mayor on June 27, 2012



Team at work – Source: Boston Globe article

Smarter Cities Challenge Boston – Thank You and Congratulations !

Jennifer Crozier to: Jane Xu, Biplav Srivastava, Raymond Rudy, Alberto Giacomel, Brent Miller, Steven Wysmuller, Cathleen Finn, Radha Ratnaparkhi, Gyana R Parija, Tsuyoshi Ide, Mauro Giraldo, Isabel Y Yang, Robert O'herron, Doris B Gonzalez

07/02/2012 10:56 PM

Cc: Ann Cramer, Lyell E Sakaue, Sophia Tu, Bijan Davari, Raghuram Krishnapuram, Kohichi Takeda, Marco Rossi, Mary Coucher, Dave Lowrie

Custom expiration date: 07/02/2042

Team Boston,
I wanted to thank you for your efforts and congratulate you on a highly successful Smarter Cities Challenge engagement. I understand that Mayor Menino was very complimentary of your efforts and is eager to move forward with your recommendations, along with terrific work preparing the local team for success and securing press coverage from the Boston Globe, NPR, and a range of local TV stations on your work. Programs like the Smarter Cities Challenge can be successful only because of the dedication , professionalism and ingenuity of the IBMers on the team, and you all are clearly among our best and brightest. Thank you again for your hard work, and I hope you enjoyed your time in Boston!

To each of the managers of Team Boston,
I would also like to thank you and your teams again for contributing your top talent to this important initiative. Their success would not be possible without the strong support of management and a team that's willing to fill in for them while they are away. They have been an amazing group ---- bringing a diverse set of skills and integrating them quickly for the Mayor and his team . I hope you will give each of them an opportunity to share this work, and the recommendations they delivered, with your team when they return.

I know there's much more to come --- eager to see the final report and action from the city ---- but I wanted to take a moment to say thank you and congratulations to an incredible team!

Jen

Jen Crozier
Vice President, IBM Global Citizenship Initiatives
1 New Orchard Road, Armonk, NY

Assistant: Tanya Mitchell

Press on the IBM SCC Boston team work:

1. Boston Globe, June 29, 2012
http://www.boston.com/business/technology/articles/2012/06/29/ibm_gives_advice_on_how_to_fix_boston_traffic__first_get_an_app/
(Alternative:
http://bostonglobe.com/business/2012/06/28/ibm-gives-advice-how-fix-boston-traffic-first-get-app/goxK84cWB9utHQogpsbd1N/story.html)

2. Popular Science, 2 July 2012
http://www.popsci.com/technology/article/2012-07/bostons-ibm-built-traffic-app-merges-multiple-data-streams-predict-ease-congestion

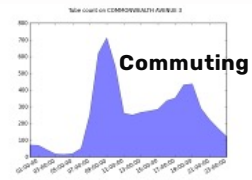
3. Others: National Public Radio (USA), and a range of local TV stations on the work.

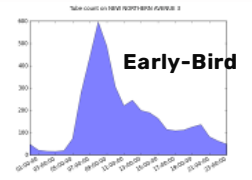
Result 3: Birds-Eye View of City Traffic from Aggregated Data

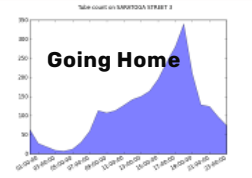
Boston traffic patterns - 24 hour time lapse



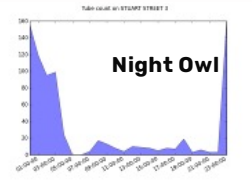
Result 2: Street Classification Based on Traffic Volume

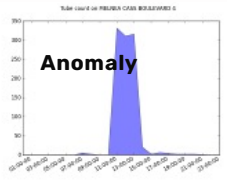


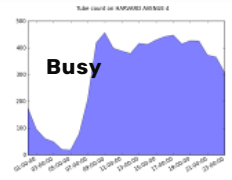












- Assign different traffic light patterns for different streets, times
- Schedule public works projects to minimize traffic impact
- Detect changes in traffic patterns to drive policy changes (parking, lanes, street)
- Assess traffic impact of new landmarks
- Inform businesses, developers

Traffic Patterns to Clusters



Average traffic volume on weekdays at 6:00 a.m.

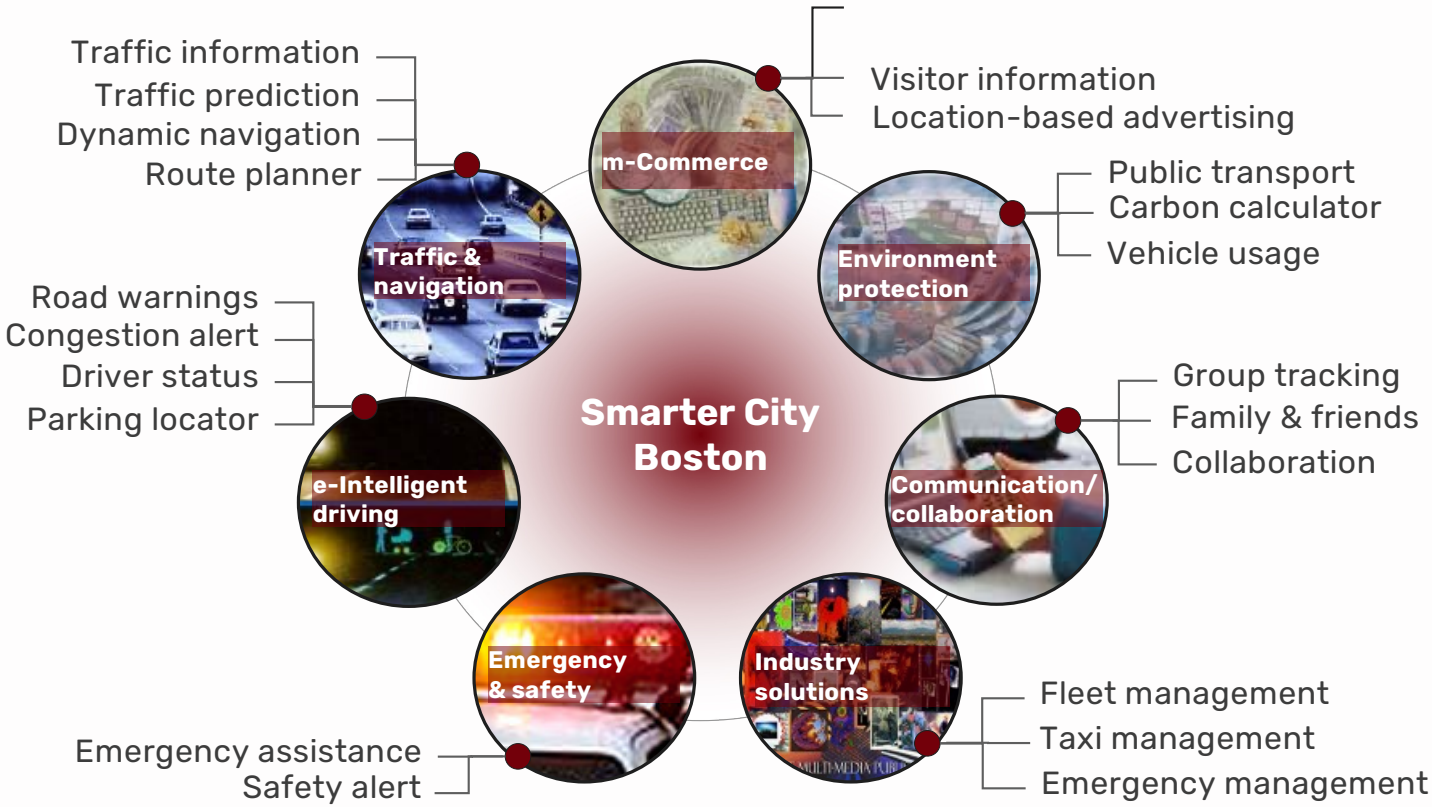


Average traffic volume on weekdays at 12:00 p.m.



All roads are not the same, and this matters for the community !

Smarter Transportation Leads to a Smarter Boston

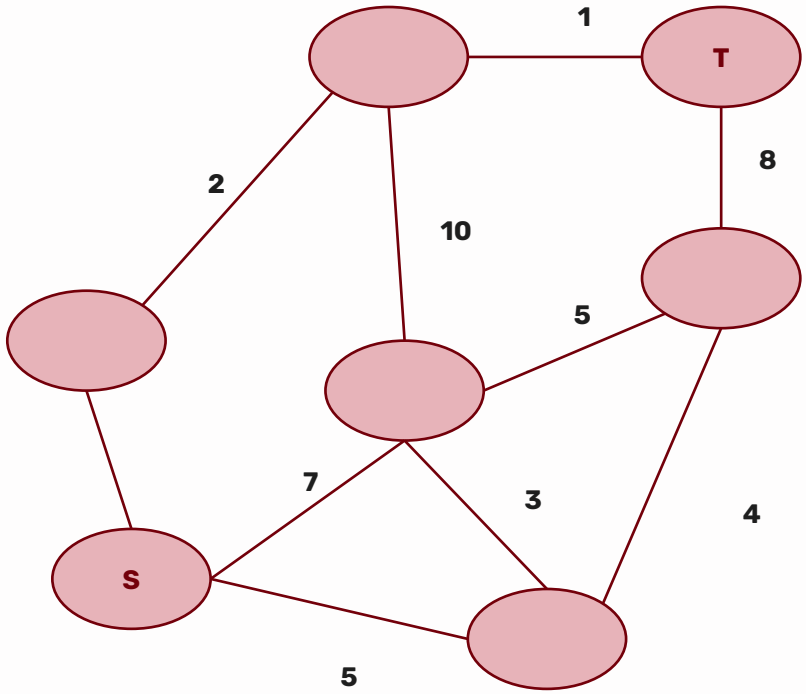


02.2

Analysis for a Specific Technical Goal – Journey Planning

Route Planning: Basics

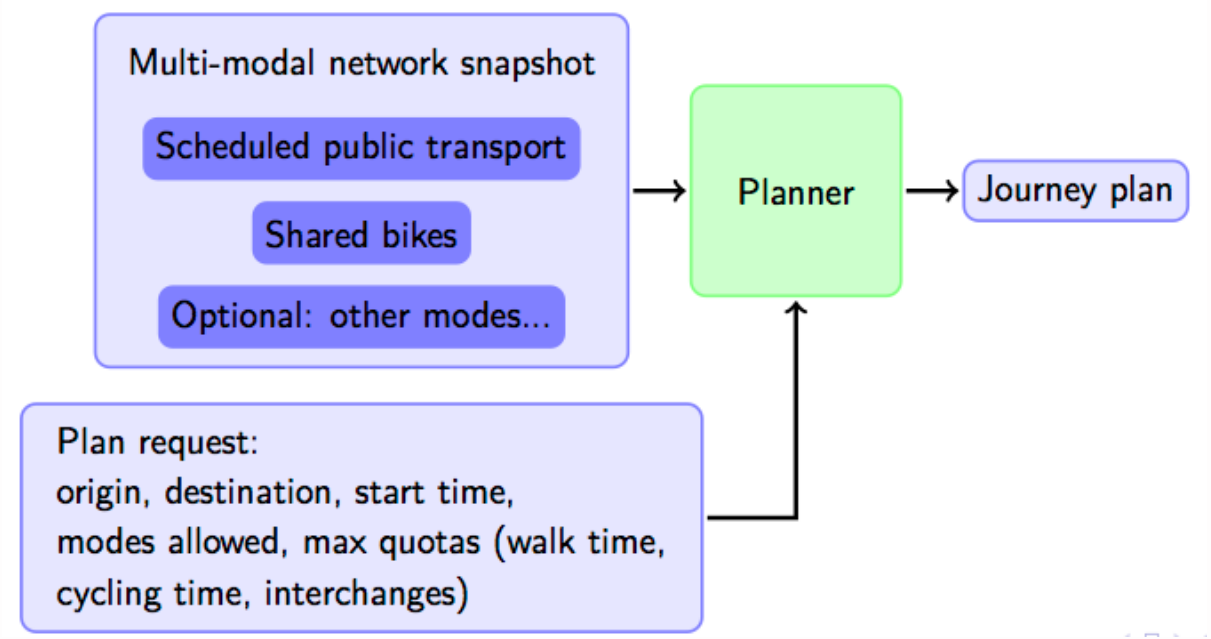
- Given a graph G=(V,E)
- Source **S**, destination **T**
- Edge length
- **Goal**: Find shortest path between **S** and **T**
- Dijkstra's algorithm



Characteristics of Route Planning

	Stochastic	Adaptive	Time Dependent Uncertainty	Deadline	Generalized Cost Function
Stochastic Route Planning	Y	N	N	Y	Y
Canadian Traveler Problem	Y	Y	N	N	N
Multimodal Route Planning	Y	Y	Y	Y	N
Stochastic OPs	Y	N	Y	Y	Y

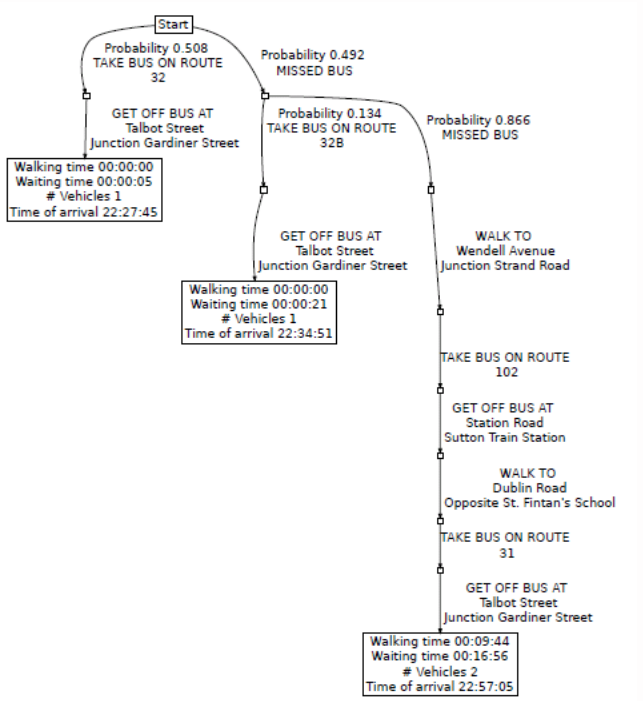
Problem Setting



Docit: An Integrated System for Risk-Averse Multi-Modal Journey Advising, Adi Botea, Michele Berlingerio, Stefano Braghin Eric Bouillet, Francesco Calabrese, Bei Chen Yiannis Gkoufas, Rahul Nair, Tim Nonner, Marco Laumanns, IBM Technical Report, 2014

A Flexible Journey Plan to Commuters to Reach Destination in All Eventuality

Pilots completed in Dublin, Ireland and Montplier, France





Docit: An Integrated System for Risk-Averse Multi-Modal Journey Advising, Adi Botea, Michele Berlingerio, Stefano Braghin Eric Bouillet, Francesco Calabrese, Bei Chen Yiannis Gkoufas, Rahul Nair, Tim Nonner, Marco Laumanns, IBM Technical Report, 2014

02.3

Using Multiple Analyses Together for Business Goal - Improving Traffic Safety

Towards Enhancing Road Safety in South Carolina Using Insights from Traffic and Driver-Education Data

Gupta, N., Muppasani, B., Srivastava, S., Goel, A., Hartfield, R., Buehrig, T., … Vasconcelos, L. (2025). *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(28), 29385–29387. https://doi.org/10.1609/aaai.v39i28.35260

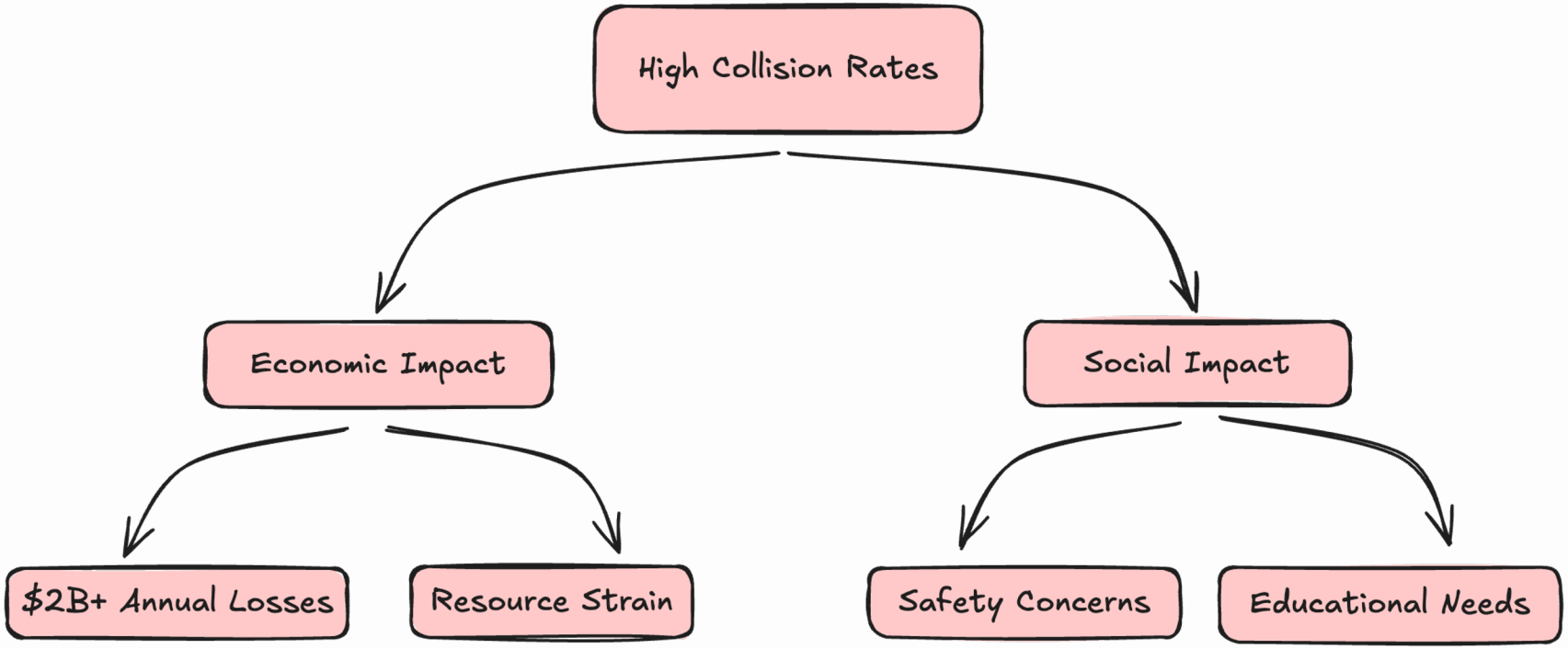
Github page: github.com/ai4society/Traffic-Data-Analysis

Project page: ai4society.github.io/projects/traffic-page

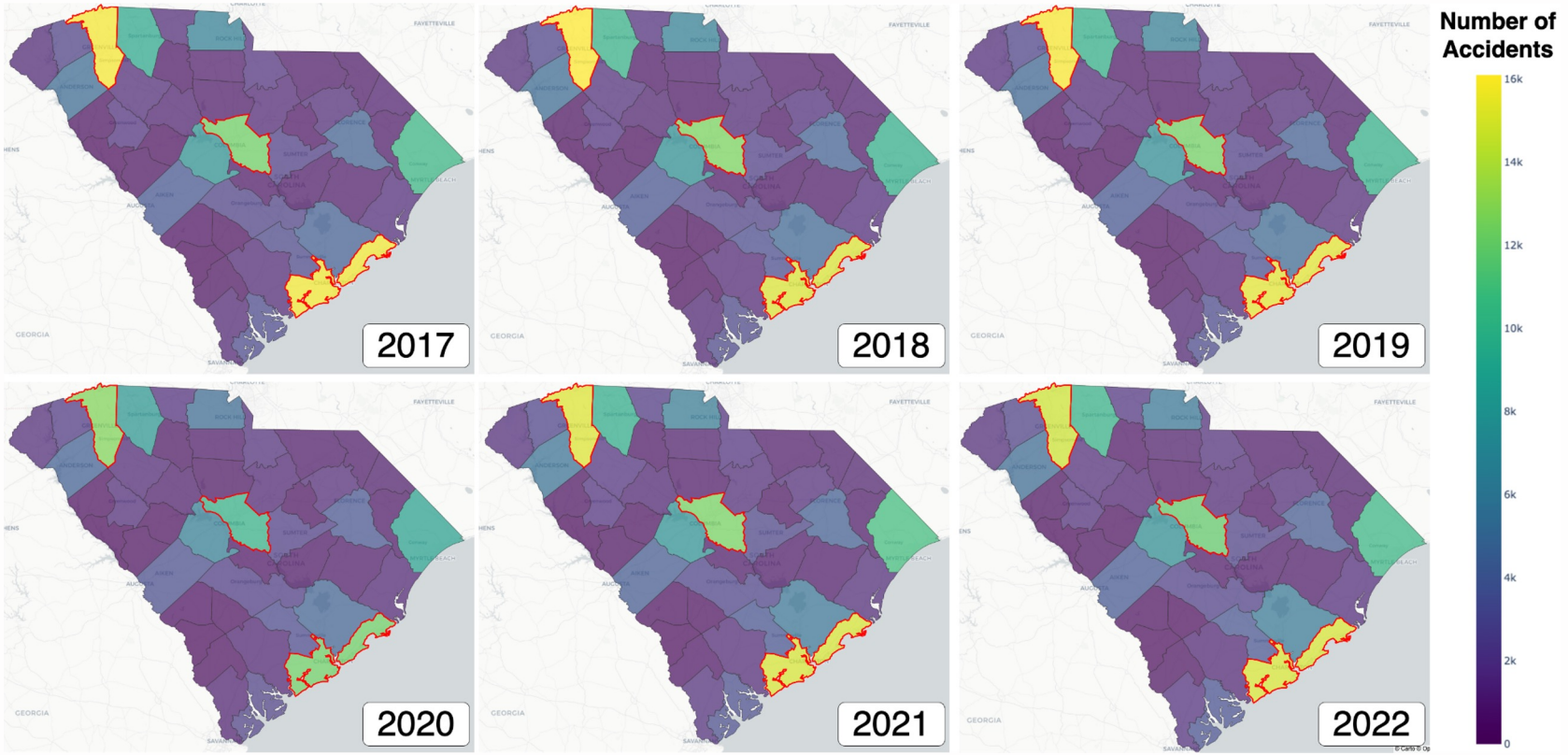
02.3 • TRAFFIC SAFETY

Understanding Traffic to Improve Driver Safety

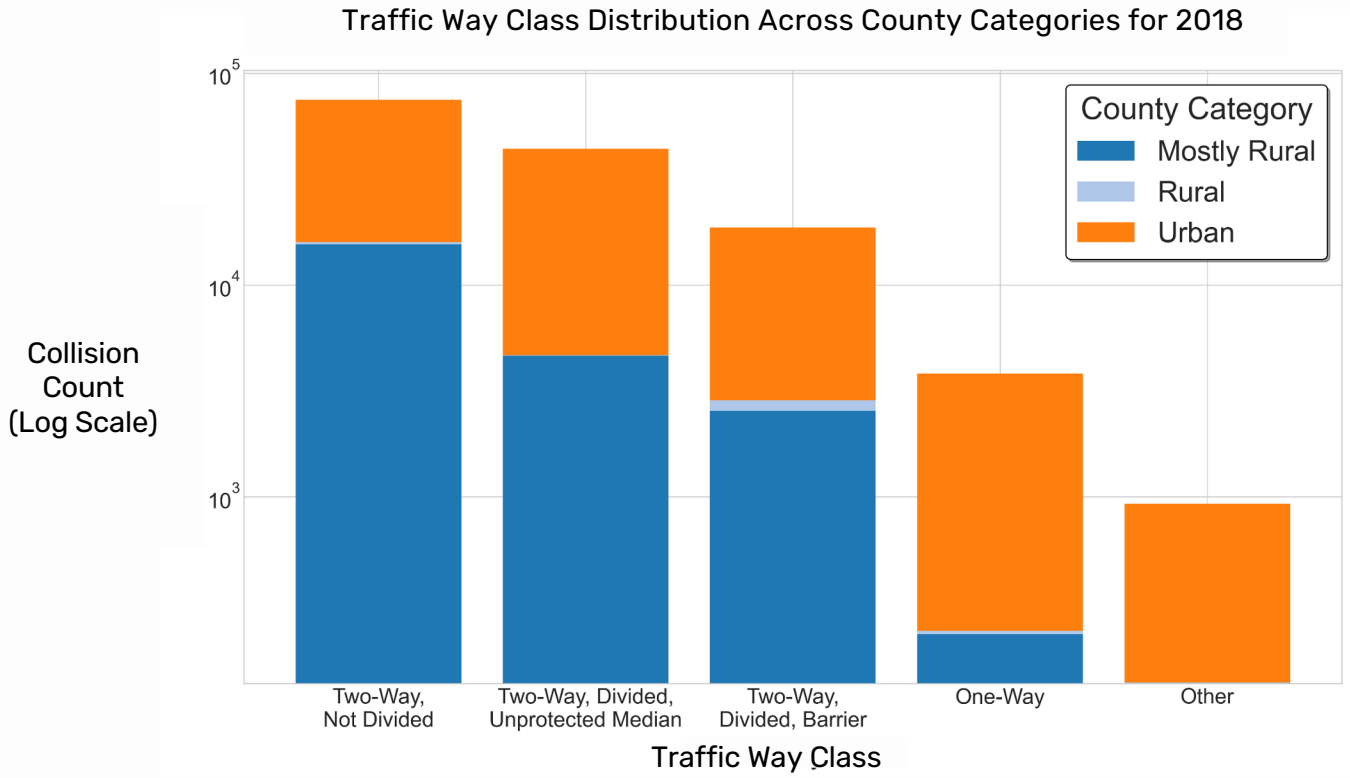
South Carolina faces a critical road safety crisis, with one of the highest traffic fatality rates in the U.S.



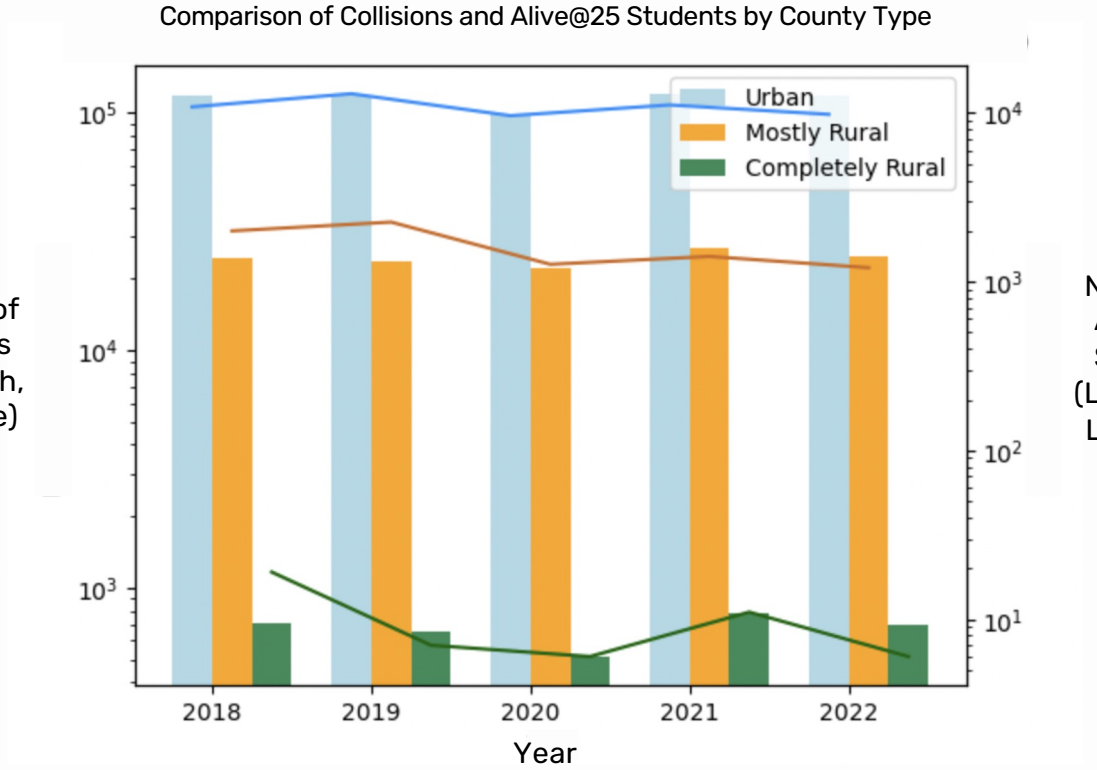
Key Findings: Geographic



Key Findings: Infrastructural

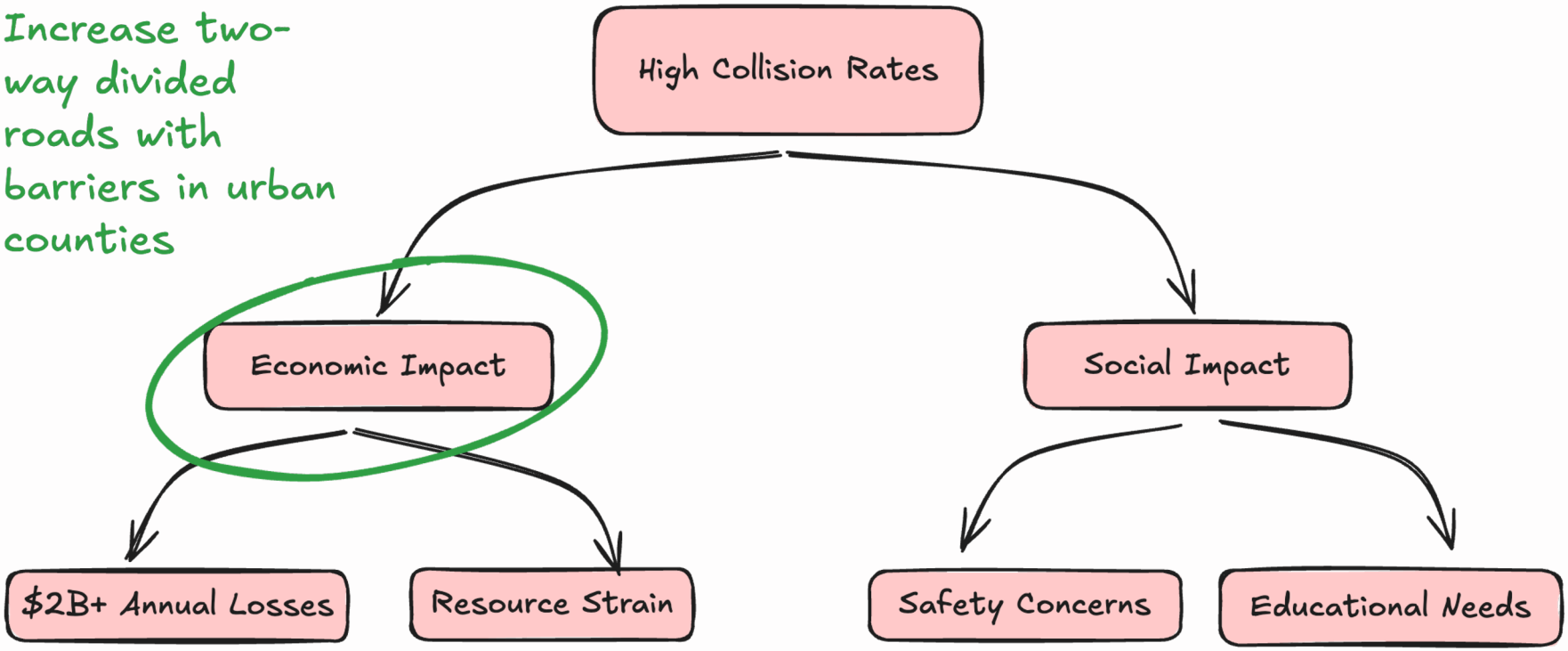


Key Findings: Educational



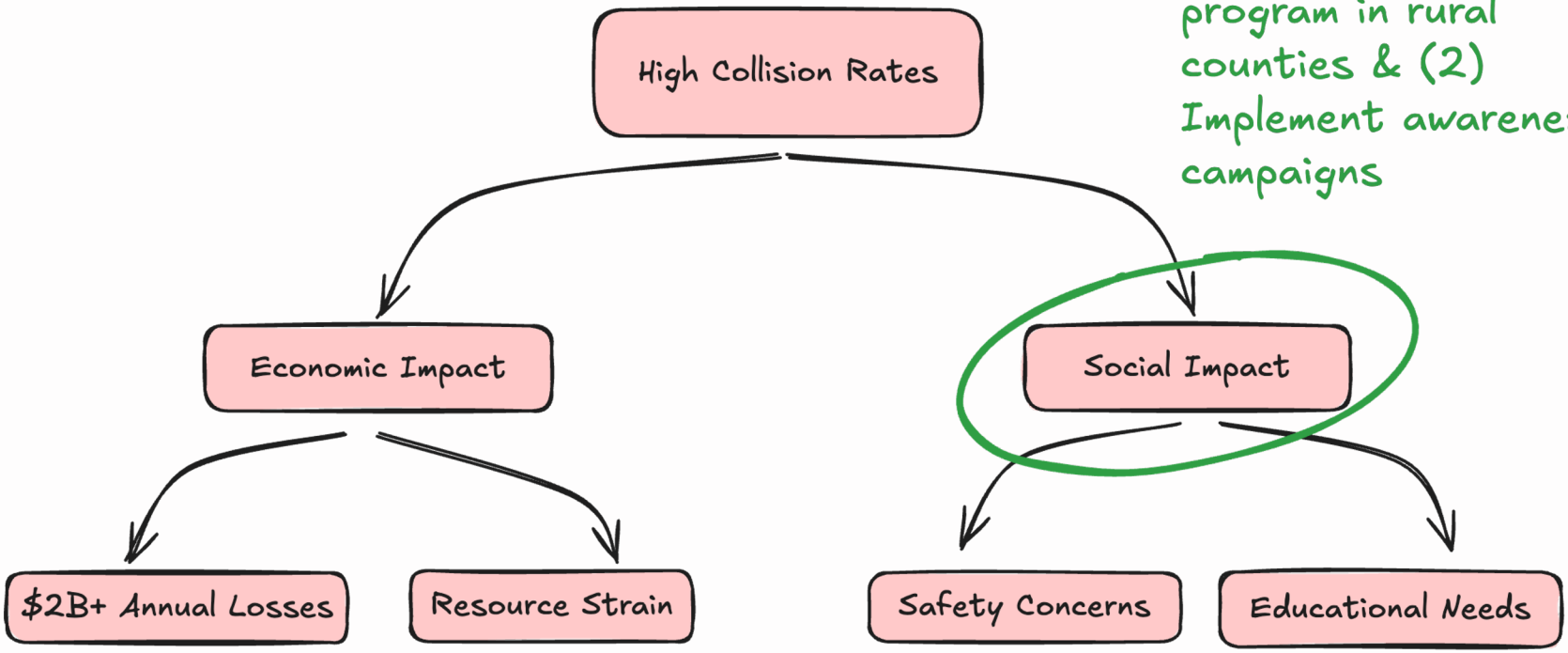
1st Recommendation

Increase two-way divided roads with barriers in urban counties



2nd Recommendation

(1) Expand Alive@25 program in rural counties & (2) Implement awareness campaigns



03

Lessons

03 • LESSONS

Lessons Summary

01 Take a multi-stakeholder approach.

02 Understand the incentives of different stakeholders involved.

03 Discover and prioritize value to people that matter.

04 Use and contribute to open traffic data.

05 Amplify solutions rather than problems to sponsors.

06 Incorporate capability building and skilling.

03 • LESSONS

Take a Multi-Stakeholder Approach

- Involves multiple stakeholders: at demand side (commuters), supply side, and policy making side of the ecosystem.
- For solving a problem effectively, their perspectives must be taken into account together.

Demand is good

Supply impact economy, quality of service

Policy ensures sustainability of solutions

03 • LESSONS

Understand the Incentives of Different Stakeholders Involved

> People and organizations respond to incentives, **but they may not be explicit about them**.

> Mis-alignment can be detected and the need for **noncomputational interventions** can be discovered.

03 • LESSONS

Discover and Prioritize Value to People That Matter

Some stakeholders are more vociferous or influential than others for solution adoption.

- For example, for traffic, commuters or car owners may be more important to the sponsor.

At least one important stakeholder should be satisfied with new solution so that they in turn become champions of the initiative.

If one tries to please all stakeholders, they often end up with having no champion, and ***the project ends unrealized.***

03 • LESSONS

Use and Contribute to Open (Traffic) Data

> Be part of the solution by releasing data using best practices

03 • LESSONS

Amplify Solutions Rather Than Problems to Sponsors

> Sponsors give problems and funding to engineers for solving.

> However, better understanding of problems does not mean sponsor were clueless. **Best way to highlight progress is with solutions**, *not just more information about the problem*.

> When sponsors do not see solution path from interim project reports, and instead see more problems or same problem but much more frequently, they can become very sensitive about proceeding further.

> Task prioritization and communication should be carefully undertaken with active participation of sponsors.

03 • LESSONS

Incorporate Capability Building and Skilling

> Stakeholders should be comfortable owning the new solution.

> Need capacity building and skilling to maintain new solutions; otherwise, this creates friction.

03 • LESSONS

Lessons Summary

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04

Discussion

04 • DISCUSSION

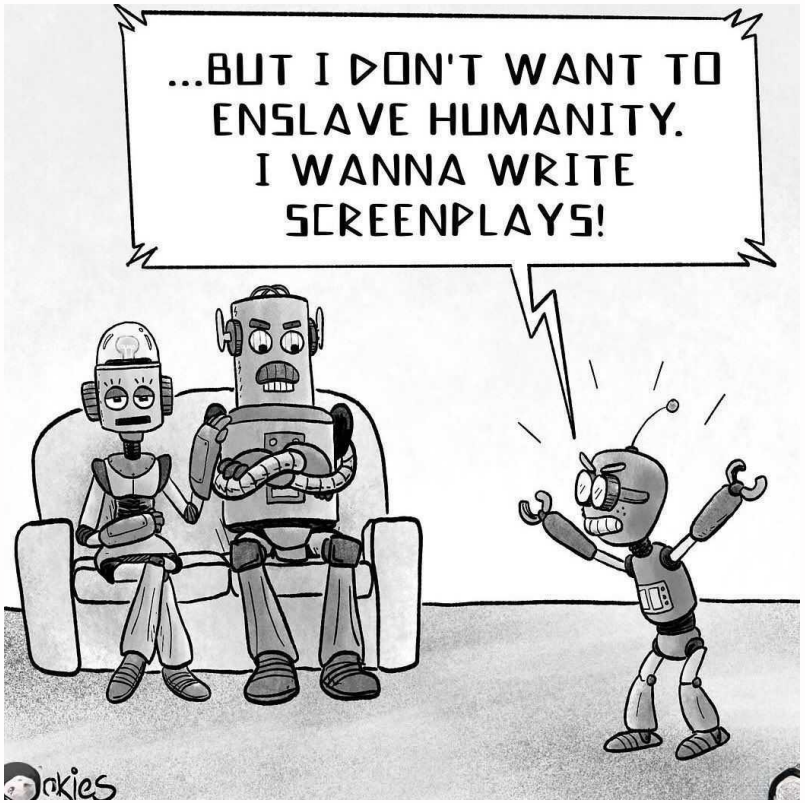
Recent Discussion – AI and Safety

"Researchers have long warned that three conditions could lead to loss of control:

- a misaligned goal,
- the capability to pursue it, and
- an environment that allows it,"

ACM A.M. Turing Award laureate Yoshua Bengio said at UN (Sep 2026)

https://thehill.com/policy/technology/6102955-un-ai-panel-urges-safeguards/



04 • DISCUSSION

Engage End Users

Open Social Innovation is an effective way to transform communities

Focus on value, data and access

In technology adoption involving people, there is no replacement for evaluation and end-user validation





- **Details**: https://tinyurl.com/safe-ai-2026
- Covered by **South Carolina Public Radio**, highlighting the workshop's panels on cybersecurity and scams, AI and mental health, and data privacy for adults 55 and older (media coverage).
- Covered by **WSPA 7**, highlighting the workshop's panels on cybersecurity and scams (media coverage(print+video)).